Orbit Constraints in Symmetric Arrays and Circulant Weighing Matrices

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Abstract

We establish two results in combinatorial design theory. For every $k \geq 3$, the minimum number of rows in a binary strength-two orthogonal array invariant under all coordinate permutations is $4\binom{k-2}{\lfloor (k-2)/2 \rfloor}$, proving Karunanayaka's Catalan-bound tightness conjecture. We also classify all minimizing shell multiplicity vectors. Separately, we prove that a circulant weighing matrix $CW(195, 81)$ does not exist, resolving a parameter pair listed as open in Arasu–Gordon–Zhang's table. Both proofs are analytic. The first combines indivisible shell masses with two moment constraints. The second combines contraction, established cyclotomic multiplier theory, a kernel character, and a short augmentation obstruction. Exact finite checks are supplied as independent corroboration, not as premises of either proof.

1 Scope and relation to the proposed method

The purpose of this note is to test a proof strategy outside periodic Dirichlet-series questions: compress constraints by symmetry, retain the integrality and coefficient restrictions that survive compression, and use them to force a sharp bound or a contradiction. The results below concern different mathematical objects and do not assume any claim about an Erdős function or an Okada lattice.

The first target is Conjecture 1 of Karunanayaka [1], whose published statement covers every $k \geq 3$ and whose reported finite verification covers $k \leq 14$. We prove the original statement with its full coordinate-permutation symmetry assumption. The second target is the existence of $CW(195, 81)$, listed in Table 10 of Arasu–Gordon–Zhang [2].

The tools used here have established precedents. Cauchy–Schwarz and orbit compression are standard. The cyclotomic multiplier theorem is due to Tan [3]; kernel-character reductions also appear in Tan [4]. The contributions asserted in this draft are the specific lower-bound proof and equality classification, and the elimination of the parameter pair $(195, 81)$. No new general multiplier theorem or universal 2-adic decision procedure is asserted.

Review status. The proofs have undergone separate internal mathematical audits and exact arithmetic checks. This is an AI-assisted research draft, not an externally refereed publication. The source check on 24 October 2026 found the targets stated as open in the cited records and found no later resolution of these specific statements. That search is not a guarantee of publication priority.

2 The Catalan bound for fully symmetric binary arrays

A binary array is *fully symmetric* if its multiset of rows is invariant under every permutation of the coordinates. Equivalently, there are nonnegative integers $m_0, \dots, m_k$ such that every binary word of weight w occurs exactly m_w times. Repeated rows are allowed. Karunanayaka [1] constructs fully symmetric strength-two orthogonal arrays with

$$B_k = 4\binom{k-2}{\lfloor (k-2)/2 \rfloor}$$

rows and conjectures that this is the minimum for every $k \geq 3$. We prove this lower bound and determine all equality cases.

Theorem 2.1. *Every nonempty fully symmetric binary orthogonal array of strength two on $k \geq 3$ coordinates has at least B_k rows. Equality holds precisely for the following multiplicity vectors, with all unlisted entries zero:*

1. *If $k = 2r+1$ is odd, either $m_{r+1} = 1$, $m_0 = C_r$, or its level flip $m_r = 1$, $m_k = C_r$.*
2. *If $k = 2r \geq 6$ is even, $m_r = 1$ and $m_0 = m_k = C_{r-1}$.*
3. *If $k = 4$, the two vectors are $(1,0,1,0,1)$ and $(0,1,0,1,0)$.*

Here $C_r = \binom{2r}{r}/(r+1)$ is the rth Catalan number.

The proof uses two moments and the fact that a populated shell contains at least one full coordinate-permutation orbit.

Lemma 2.2 (A forced-shell bound)**.** *Let a multiset of $N > 0$ real numbers have sum zero and square-sum kN, with $k > 0$. Suppose M of its entries are equal to d. Then*

$$N \geq M\left(1 + \frac{d^2}{k}\right).$$

If also $d = 0$ and every entry has absolute value at most $k > 1$, then $N \geq kM/(k-1)$.

Proof. The remaining $N - M$ entries have sum $-Md$ and square-sum $kN - Md^2$. Cauchy–Schwarz gives

$$M^2 d^2 \leq (N-M)(kN - Md^2),$$

which simplifies to the first inequality. For the second, $kN \leq k^2(N-M)$. □

Lemma 2.3. *For $k \geq 3$ and nonzero integers d with $d \equiv k \pmod 2$ and $d^2 \leq k$,*

$$\binom{k}{(k-|d|)/2}\left(1 + \frac{d^2}{k}\right) \geq B_k.$$

If k is odd, equality holds only for $|d| = 1$. If k is even, equality holds only when $k = 4$ and $|d| = 2$.

Proof. For nonnegative admissible d put $G_k(d) = \binom{k}{(k-d)/2}(k + d^2)$. The ratio of consecutive values is

$$\frac{G_k(d+2)}{G_k(d)} = \frac{k-d}{k+d+2}\,\frac{k + (d+2)^2}{k + d^2}.$$

The difference of cross products is

$$(k-d)(k+(d+2)^2)-(k+d+2)(k+d^2)=2(d+1)(k+1-(d+1)^2).$$

This is positive whenever $(d+2)^2 \le k$, so G_k strictly increases along the admissible range. For odd k, its least nonzero argument is 1, and

$$\frac{G_k(1)}{k}=\frac{k+1}{k}\binom{k}{(k-1)/2}=B_k.$$

For even k, the least nonzero argument is 2. With $C=\binom{k}{k/2}$, we have

$$\frac{G_k(2)}{k}=C\frac{k+4}{k+2}\ge C\frac{k}{k-1}=B_k,$$

where the difference of the two rational factors is $(k-4)/((k+2)(k-1))$. This also gives the equality statements. □

Proof of Theorem 2.1. Replace each binary coordinate x_i by $1-2x_i$, and let D denote the sum of the k signs in a row. Strength two gives

$$\sum D=0, \qquad \sum D^2=kN.$$

In the shell of weight w, the value of D is $d_w=k-2w$ and the number of rows is $M_w=m_w\binom{k}{w}$. Since the average of D^2 is k, some populated shell has $d_w^2\le k$. Write $d=d_w$ and $M=M_w$ for such a shell. Its positive integer multiplicity gives

$$M\ge\binom{k}{(k-|d|)/2}.$$

If $d\neq 0$, Lemmas 2.2 and 2.3 imply $N\ge B_k$. If $d=0$, then k is even and Lemma 2.2 gives

$$N\ge\frac{k}{k-1}M\ge\frac{k}{k-1}\binom{k}{k/2}=B_k.$$

This covers the boundary $d^2=k$ as well.

For equality at odd k, Lemma 2.3 forces $|d|=1$, and the selected shell has multiplicity one. Equality in Cauchy–Schwarz makes all remaining values of D equal. Their mean is $-k/d$, so they are all the opposite endpoint $\mp k$. The endpoint multiplicity is

$$\frac{1}{k}\binom{k}{(k-1)/2}=C_{(k-1)/2},$$

giving the two displayed vectors.

For even $k\ge 6$, the nonzero-d bound is strict, so equality requires a central shell of multiplicity one. Equality in $\sum D^2\le k^2(N-M)$ makes all remaining rows endpoints. Zero mean gives equal endpoint multiplicities, each

$$\frac{1}{2(k-1)}\binom{k}{k/2}=C_{k/2-1}.$$

For $k=4$ this still gives $(1,0,1,0,1)$; equality with $|d|=2$ is also possible, and Cauchy–Schwarz then forces the opposite shell of weight 3 or 1, giving $(0,1,0,1,0)$.

Finally, every displayed vector satisfies $\sum D=0$ and $\sum D^2=kN$. Coordinate symmetry turns these two identities into zero first-order sign means and zero pairwise sign correlations, hence strength two. Their row counts are B_k, proving attainability and the complete classification. □

Remark 2.4. The upper-bound constructions are those of Karunanayaka [1]. The lower bound uses integrality only through the implication $m_w > 0 \Rightarrow m_w \geq 1$. This is a theorem for full coordinate-permutation symmetry, not for arbitrary binary orthogonal arrays.

Remark 2.5. For $k = 15$, the minimum is $B_{15} = 6864$. The two minimizers have $m_8 = 1$, $m_0 = 429$, and their level flip $m_7 = 1$, $m_{15} = 429$; indeed $\binom{15}{8} + 429 = 6435 + 429 = 6864$. For $k = 16$, the minimum is $B_{16} = 13728$, and the unique minimizer has $m_8 = 1$, $m_0 = m_{16} = 429$, since $\binom{16}{8} + 2 \cdot 429 = 12870 + 858 = 13728$. These are the first two values beyond the range $k \leq 14$ verified in [1].

3 Nonexistence of a circulant weighing matrix of order 195 and weight 81

A circulant weighing matrix $CW(v, k)$ is equivalent to an element $A \in \mathbb{Z}[C_v]$ whose coefficients belong to $\{0, 1, -1\}$ and which satisfies $AA^{(-1)} = k$. Here the involution combines inversion of group elements with complex conjugation of coefficients. The augmentation $\epsilon(A)$ is the sum of the coefficients. We prove the following parameter-specific nonexistence result.

Theorem 3.1. *There is no circulant weighing matrix* $CW(195, 81)$.

The pair $(195, 81)$ appears among the unresolved cases in Arasu–Gordon–Zhang, *New nonexistence results on circulant weighing matrices*, Table 10, arXiv:1908.08447v3 (2021). Their Table 9 records two equivalence classes of $ICW_3(65, 81)$. Tan's September 2026 preprint, *Character and Multiplier Obstructions for Circulant Weighing Matrices*, arXiv:2608.18468v2, resolves eight other entries. Its methods include the same general kernel-character reduction employed below; neither that reduction nor the multiplier theorem is claimed as new here.

We use the integral multiplier theorem and the following cyclotomic version, proved in Tan, *Group Invariant Weighing Matrices*, arXiv:1610.01914v4 (2018), Theorem 3.4 and Corollary 3.5(b). If $X \in \mathbb{Z}[\zeta_u][G]$, where G is finite abelian,

$$XX^{(-1)} = n, \qquad (n, |G|) = 1,$$

and the cyclotomic automorphism σ_t fixes every prime ideal above the prime divisors of n in $\mathbb{Q}(\zeta_{\mathrm{lcm}(u, \exp G)})$, then $X^{(t)} = \eta g X$ for a root of unity $\eta \in \mathbb{Z}[\zeta_u]$ and $g \in G$. If additionally $(t - 1, \exp G) = 1$, a group translation removes g. In $X^{(t)}$, the coefficients are acted on by σ_t and the group elements are raised to their tth powers.

Lemma 3.2. *Every* $B \in \mathbb{Z}[C_{65}]$ *satisfying*

$$BB^{(-1)} = 81, \qquad \epsilon(B) = 9, \qquad |B_x| \leq 3$$

has coefficients in $\{0, 3, -3\}$.

Proof. Apply Tan's generalized multiplier theorem with coefficient conductor $u = 1$, group C_{65}, norm $n = 81$, and multiplier integer 3. Here $(81, 65) = 1$, and the automorphism $\zeta_{65} \mapsto \zeta_{65}^3$ is the Frobenius at 3, fixing every prime ideal above 3. Thus $B^{(3)} = \eta X^j B$ with $\eta \in \{1, -1\}$. Augmentation $\epsilon(B) = 9$ forces $\eta = 1$, and $(3 - 1, 65) = 1$ permits a group translation with $B^{(3)} = B$. This applies to the bounded integer contraction directly; it does not require its coefficients to be $0, \pm 1$. Use $C_{65} = C_5 \times C_{13}$. Multiplication by 3 has order 4 on $C_5 \setminus \{0\}$ and order 3 on $C_{13} \setminus \{0\}$. Its orbit coefficients can therefore be written as

$$a \quad (\text{size } 1), \qquad b \quad (\text{size } 4), \qquad c_1, \ldots, c_4 \quad (\text{size } 3), \qquad d_1, \ldots, d_4 \quad (\text{size } 12).$$

The c_i and d_i use the same ordering of the four nonzero C_{13} orbits. All ten coefficients are integers of absolute value at most 3. The energy equation is

$$a^2 + 4b^2 + 3\sum_i c_i^2 + 12\sum_i d_i^2 = 81. \tag{3.1}$$

Projecting onto C_5 gives coefficient $x = a + 3\sum_i c_i$ at zero and coefficient $y = b + 3\sum_i d_i$ at each nonzero element. Consequently

$$x + 4y = 9, \qquad x^2 + 4y^2 = 81.$$

Elimination gives $4y(5y - 18) = 0$, whence $y = 0$ and $x = 9$ by integrality. Thus $a = 3\alpha$ and $b = 3\beta$ with $\alpha, \beta \in \{-1, 0, 1\}$, and

$$\sum_i c_i = 3 - \alpha, \qquad \sum_i d_i = -\beta. \tag{3.2}$$

The projection onto C_{13} has energy

$$(a + 4b)^2 + 3\sum_i (c_i + 4d_i)^2 = 81. \tag{3.3}$$

If $\beta \neq 0$, then $|3\alpha + 12\beta| \leq 9$ forces $\alpha = -\beta$. The first term of (3.3) is then 81, so $c_i + 4d_i = 0$ for all i. Since $|c_i| \leq 3$, every $d_i = 0$, contradicting (3.2). Hence $\beta = 0$.

Set $t = \sum_i d_i^2$. Comparing (3.1) and (3.3) now yields

$$\sum_i d_i = 0, \qquad \sum_i c_i = 3 - \alpha, \qquad \sum_i c_i^2 + 4t = 27 - 3\alpha^2, \qquad 2\sum_i c_i d_i + 3t = 0. \tag{3.4}$$

For $t > 0$, Cauchy–Schwarz gives

$$\frac{9}{4}t^2 \leq t(27 - 3\alpha^2 - 4t), \qquad t \leq \frac{108}{25} < 5.$$

Also $t \equiv \sum_i d_i = 0 \pmod 2$. It remains to consider $t = 0, 2, 4$.

If $t = 4$, the d_i are, in some order, $(1, 1, -1, -1)$. Let P and N be the sums of the c_i at the positive and negative positions. Then $P - N = -6$ and $P + N = 3 - \alpha$. For $\alpha = 0$ these equations contradict integrality. For $\alpha = 1$ they give $(P, N) = (-2, 4)$, so $\sum_i c_i^2 \geq 2 + 8 = 10$, whereas (3.4) requires 8. For $\alpha = -1$ they give $(P, N) = (-1, 5)$, so $\sum_i c_i^2 \geq 1 + 13 = 14$, again exceeding 8. Thus $t \neq 4$.

If $t = 2$, arrange $d = (1, -1, 0, 0)$. Then $c_1 - c_2 = -3$, so

$$(c_1, c_2) \in \{(-3, 0), (-2, 1), (-1, 2), (0, 3)\}.$$

Their squared norms are $9, 5, 5, 9$. For $\alpha = \pm 1$, the remaining squared norm $c_3^2 + c_4^2$ must be 7 or 11, neither a sum of two squares. For $\alpha = 0$ it must be 10 or 14. The latter is impossible. In the former case the first pair is $(-3, 0)$ or $(0, 3)$, requiring $c_3 + c_4 = 6$ or 0. But the solutions to $c_3^2 + c_4^2 = 10$ have sums in $\{-4, -2, 2, 4\}$. Thus $t \neq 2$.

We conclude that $t = 0$, so all $d_i = 0$. If $\alpha = \pm 1$, then $\sum_i c_i^2 = 24$. Reduction modulo 4 forces either zero or four odd c_i. Four odd entries give squared sum $4 \pmod 8$, while zero odd entries, with $|c_i| \leq 3$, give squared sum at most 16. Both are impossible. Hence $\alpha = 0$ and $\sum_i c_i^2 = 27$. There must be three odd entries, since one odd and three even entries have squared sum at most $9 + 12 = 21$. Modulo 8, the remaining even entry must be zero. The three odd entries must then all have absolute value 3. Their sum is 3, so two equal 3 and one equals -3. This proves the stated divisibility of every coefficient of B. □

Proof of Theorem 3.1. Suppose $A \in \mathbb{Z}[C_{195}]$ represents such a matrix, with $\epsilon(A) = 9$. Write $C_{195} = C_{65} \times C_3$ and contract the C_3 coordinate. The result is an integer contraction B satisfying Lemma 3.2; hence $B = 3C$ with C having coefficients $0, \pm 1$.

Let ω be a primitive cube root of unity and put $\lambda = 1 - \omega$, so $\lambda\overline{\lambda} = 3$. Apply the nonprincipal character $Y \mapsto \omega$ of C_3 to A, obtaining $E \in \mathbb{Z}[\omega][C_{65}]$ with $EE^{(-1)} = 81$. Each three-entry fiber has principal sum 0 or ± 3. A fiber of sum ± 3 is constant and has nonprincipal image zero. A nonzero zero-sum fiber is a permutation of $(1, -1, 0)$ and has image λ times an Eisenstein unit. Thus

$$E = \lambda D, \qquad DD^{(-1)} = 27, \qquad D_x \in \{0, \pm 1, \pm\omega, \pm\omega^2\}. \tag{3.5}$$

In particular, D has precisely 27 nonzero coefficients.

Apply the generalized multiplier theorem with coefficient conductor 3, group C_{65}, norm 27, and multiplier integer 133. We have $(27, 65) = 1$, $(133, 195) = 1$, and

$$133 \equiv 1 \pmod 3, \qquad 133 \equiv 3 \pmod{65}.$$

The automorphism σ_{133} fixes every prime above 3 in $\mathbb{Q}(\zeta_{195})$: its restriction to $\mathbb{Q}(\zeta_{65})$ is the Frobenius automorphism at 3, and adjoining ω is totally ramified at these primes, so each has a unique extension. The theorem therefore gives $D^{(133)} = \eta X^j D$. Since σ_{133} is the identity on $\mathbb{Z}[\omega]$, augmentation gives $\epsilon(D) = \eta\epsilon(D)$. By (3.5), $|\epsilon(D)|^2 = 27$, so $\eta = 1$. Finally $(133 - 1, 65) = 1$, allowing a group translation after which $D^{(133)} = D$. Equivalently, D is constant on multiplication-by-3 orbits of C_{65}.

These orbits have sizes $1, 4$, four copies of 3, and four copies of 12. The support size 27, reduced modulo 3, forces the size-1 and size-4 orbits both to be absent. The remaining support equation is

$$12k + 3l = 27, \qquad 0 \le k, l \le 4,$$

whose only solution is $(k, l) = (2, 1)$. Consequently

$$\epsilon(D) = 12u + 12v + 3w = 3(4u + 4v + w)$$

for Eisenstein units u, v, w. Its norm must be 27, so $|4u + 4v + w| = \sqrt{3}$. If $v = -u$, this absolute value is 1. Otherwise $u + v$ is a nonzero Eisenstein integer, and therefore $|u + v| \ge 1$. The reverse triangle inequality then gives

$$|4u + 4v + w| \ge 4|u + v| - |w| \ge 3.$$

Neither alternative equals $\sqrt{3}$. This contradiction proves the theorem. □

Reproducibility and limits of the transfer. The proof is entirely analytic. Independent standard-library calculations enumerate 3388 contraction candidates satisfying augmentation and energy, recovering eight normalized contractions in the two previously recorded equivalence classes. A separate exact calculation checks all 864 normalized Eisenstein orbit assignments; four survive the first correlation equation and none survive the second. These checks are redundant certificates rather than hypotheses of the proof.

The successful transfer is the retention of character information inside fibers, followed by a bounded-coefficient and augmentation obstruction. It is not a general solution of the circulant Hadamard problem. In fact, for every odd u, its order-4 quotient at candidate order $4u^2$ has the feasible vector $(u, u, u, -u)$. This vector has the correct parity, coefficient bounds, and flat quotient Fourier norm, and each entry lifts separately to a sum of u^2 signs. Thus a test using only the maximal 2-power quotient and those constraints cannot exclude any such candidate order.

4 Verification and limits

4.1 Checks supplied with this note

The general proofs above do not rely on finite search. The supplementary programs use exact arithmetic to check the arithmetic reductions independently. For the array result, two implementations verify 10,813 shell inequalities through $k = 1000$, including the boundary $d^2 = k$, and check the constructions against the original one- and two-coordinate marginal equations. A separate small-parameter enumeration reproduces all minimum multiplicity vectors; an optional CP-SAT run checks $3 \le k \le 16$.

For the circulant result, an exhaustive orbit calculation recovers eight normalized contractions in two equivalence classes. An independent Eisenstein calculation excludes the normalized unit-orbit assignments. These numerical checks are unnecessary for the final proof: the contraction lemma and the augmentation inequality already exclude every possible matrix.

4.2 What transfers, and what does not

For the array problem, positivity makes a populated orbit contribute at least its full cardinality. The moment equations then force enough mass outside it. There is no need for a 2-adic obstruction. For the circulant problem, a quotient alone discards information: evaluating a nonprincipal character of its kernel retains enough information to force a unit-valued group-ring equation. The resulting orbit sizes conflict with the permitted augmentation norm.

This does not justify a general implication from local integral feasibility to a solution with a prescribed finite alphabet or with nonnegative coefficients. Graph-flow, marginal-table, tiling, and Littlewood-polynomial targets were considered separately; no resolution of their broad open conjectures is asserted here. In particular, neither result proves the Hadamard conjecture, the circulant Hadamard conjecture, or Tutte's flow conjectures. The evidence for transfer consists of the two precise theorems proved above, with the hypotheses stated there.

References

[1] R. C. Karunanayaka, *Catalan Bounds for Symmetric Strength-Two Orthogonal Arrays*, Journal of Combinatorial Designs, first published 18 September 2026. doi:10.1002/jcd.70039. Conjecture 1 and Theorem 6.

[2] K. T. Arasu, D. M. Gordon, and Y. Zhang, *New nonexistence results on circulant weighing matrices*, Cryptography and Communications **13** (2021), 775–789. arXiv:1908.08447v3. Tables 9 and 10.

[3] M. M. Tan, *Group Invariant Weighing Matrices*, Designs, Codes and Cryptography **86** (2018), 2677–2702. arXiv:1610.01914v4. Theorem 3.4 and Corollary 3.5(b).

[4] M. M. Tan, *Character and Multiplier Obstructions for Circulant Weighing Matrices*, preprint, 1 September 2026. arXiv:2608.18468v2.